THE MARCY LAB SCHOOL

Structured vs. Unstructured Data





Module 0: Foundations

By the End of Today, You'll Be Able To…

1

Explain what makes data structured or not

2

Classify a real file, with a reason

CASE STUDY

Star Ratings Looked Fine. Reviews Said Otherwise

4.2 rating

Ratings held steady for months

Reviews

Written comments already named the bug

Discussion

What would it take to actually use the reviews, not just the stars?

Illustrative scenario for class discussion, not a real company

A Box Score vs. a Post-Game Interview

Box score: structured

Same columns for every player.

Player	PTS	REB	AST
Jordan	24	8	5
Taylor	18	10	7
Desean	12	6	4

Interview: unstructured

Free-flowing answers, no fixed fields.

Reporter: "What changed tonight?"

Coach: "We adjusted after halftime."

SAME GAME • DIFFERENT DATA

Key Terms



Structured

Data with the same defined fields across records.



Unstructured

Free-form content without fixed fields.

THE CONCEPT

Code Along: Seeing It for Real



The analogy is easy to nod along to. Spotting the real difference in a file is the skill.

Look at a Structured File

```
$ head -n 3 data/energy_sample.csv
```

Every row has the same columns. That consistency is what structured means.

Look at Unstructured Text

```
$ echo "Great class today, the git demo finally
clicked for me!" > note.txt
$ touch note2.txt
$ echo ”I’m learning how to combine information!” >
note2.txt
$ cat note.txt note2.txt
```

Real information, but nothing to put in a “column” the same way every time.



Semi-Structured Data Lives in the Middle



Structured

Fixed rows and columns



Semi-structured

Some organization, but not fixed rows and columns.
Example: JSON



Unstructured

Free text, images, and audio

COMMON IN REAL DATA-ANALYST WORK



THE MIDDLE ZONE

Code Along: Pulling a Field Out



Unstructured doesn’t mean unusable. You just need a different tool for it.

Pulling a Field Out of Unstructured Text

```
$ grep "git" note.txt
```

This is a tiny preview of turning text into something searchable.



Structured vs. Unstructured, Side by Side



Structured

Fixed columns, same for every row

Easy to sort, filter, and compare

Example: energy_sample.csv



Unstructured

No fixed fields: free text, images

Rich detail, harder to compare directly

Example: customer review

IN COMMON: BOTH ARE REAL, VALID DATA



CHECK FOR UNDERSTANDING

Is a folder of scanned receipts structured or unstructured?



A

Structured

B

Unstructured

ANSWER KEY

Scanned Receipts Are Unstructured



B Unstructured

Each photo is an image with no consistent fields to compare across receipts — unless someone typed the amounts into a spreadsheet.

NAMED IT CORRECTLY

Code Along:
Row Count vs. Word Count



Structured and unstructured data don’t share the same natural unit to count.

Row Count vs. Word Count

```
$ wc -l data/energy_sample.csv
$ wc -w note.txt
```

Rows for structured files. Words for unstructured text.

The Data Type Shapes the Tool You Reach For

Tool choice

It shapes which tool you reach for

Analysis risk

A wrong call breaks the analysis later

Core skill

This module’s skill is tested directly

Case study link

Same trap as today’s case study

QUICK CHALLENGE

Name a File Type From Your Own Life

Round-robin: name one file type. The room calls it…

A

Structured

B

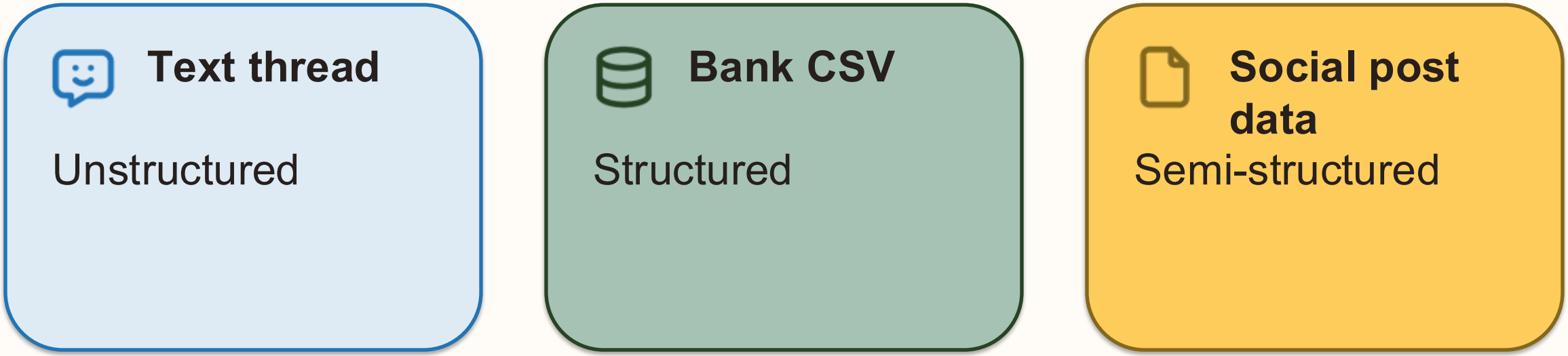
Unstructured

C

Semi-structured

ANSWER KEY

The Answer Depends on the File You Name



A live answer should always include the reason, not just the label.

WOULD YOU RATHER?

Which Would You Rather Analyze?



1,000 rows

Clean spreadsheet data



1,000 emails

Unread customer emails

DIFFERENT TOOLS • DIFFERENT QUESTIONS

THE PRACTICE

Wrapping Up



Spotting structured vs. unstructured is a judgment call you'll make on every new dataset you open.

Key Terms — Recap

Structured data

Consistent rows and columns.

Unstructured data

No fixed fields — free text, images, audio.

What You Have Now

1

A working definition of both terms

2

Practice classifying a real file

Before Next Class

Coming soon

Recall + game day

Next up

AI literacy

Practice

Column-Typing Practice
Lab, paired with AI literacy

KEEP LEARNING

Where to Go From Here



What Is Structured Data?

IBM reading. 8–10 minutes.
Clear overview with real examples.



Awesome Public Datasets

GitHub practice. 10 minutes to browse.
Spot structured vs. unstructured yourself.